

Design of CMOS Based 1-bit Comparator with MTCMOS and Forced Stack Technique in 180nm Technology

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Abstract— Leakage power and propagation delay have adverse effects on the working of CMOS(nanometer) technology-based logic circuits. The motivation behind this research work is to cut down the leakage power dissipation and propagation delay, the major influencing parameters in logic circuits. The proposed work is on one-bit comparator which uses Multi threshold technique (MTCMOS), stacking technique and gate minimising logic. The technology used is 180nm and voltage used is 1.8V. The work is performed and analyzed in SPICE. The new circuit has improved performance in power dissipation as well as propagation delay. The modified circuit shows power dissipation of 48.12pW and delay of 330pS. Comparison with previously reported circuit shows advantage of low power and less delay.

Keywords— CMOS; Delay; Leakage power; MTCMOS; Power; Stacking.

I. INTRODUCTION

Data comparison is one of the major requirements in analog and digital circuits while performing different types of operation. It comprises of two voltages (analog circuits) or two binary data (digital circuits) when either of them is compared the result gives the largest and smallest one. Digital comparator is used in various digital circuits to compare binary data. Digital comparator is used to compare two 'n' bit binary numbers say X and Y as shown in Fig.1 [1-2]. The comparator compares these two numbers by first comparing the two most significant bits (MSB). If the MSB is greater for Y then Y is obviously greater than X and circuit stops comparing other bits otherwise, check the next significant bit and so on. Let X=1001 and Y=1010 then, the circuit compares two MSB as they are equal it goes ahead and compares the next significant bit and so on and gives the

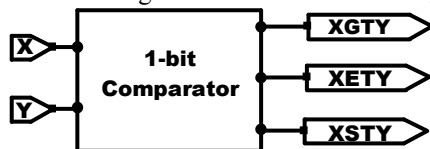
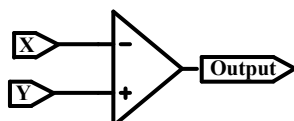


Fig.1. Block diagram of a 1-bit comparator

Fig.2. Symbol of the 1-bit comparator

Power in VLSI means the power dissipated by the circuit under its normal working. Power is dissipated in various forms in CMOS but mainly consists of two types of power.



These are static power and dynamic power. The dynamic power is due to two components, switching power and short-circuit power. Switching power is the power dissipated when transistors are in active state and inputs switch from one state to another state. Short circuit power is due to the direct path created between V_{dd} and ground. Static power is the power dissipated by the circuit when inputs are constant i.e. there are no variations in inputs [3].

TABLE I. Truth Table of 1-bit comparator

INPUTS		OUTPUTS		
X	Y	XGTY	XETY	XSTY
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

Dynamic power is the significant component in power dissipation till now. Dynamic power can be reduced by voltage scaling as power dissipation is directly proportional to the square of V_{dd} . With technology scaling, the main problem is the increase in subthreshold leakages due to a reduction in V_{th} (threshold voltage). Subthreshold current leakage depends exponentially on the threshold voltage [4].

$$P_{dynamic} = f C_L V_{dd}^2 \quad (1)$$

$$I_{sub} = A e^{k(V_{GS} - V_{th})} (1 - e^{-\frac{V_{DS}}{V_{th}}}) \quad (2)$$

With exponential relation between these, subthreshold leakage current increases as threshold voltage decreases. The static component is the major threat with a reduction in transistor sizing. Modern circuits use different technologies to reduce the static power dissipation such as stacking technique, Multi-threshold CMOS(MTCMOS), forced stack etc. Rest of the paper is described in various sections as follows:

Section II describes the various techniques to reduce the power dissipation in a circuit such as MTCMOS and forced stacking (forced stack technique in the inverter and the MTCMOS technique in NOR). Section III discusses the design of improved comparator. Section IV gives the simulation and results of the improved circuit. Section V concludes the work.

II. POWER REDUCTION TECHNIQUE

A. Transistor Stacking

Sub-threshold current is the current that passes from drain to source via the channel. A significant amount of current flows when V_{GS} is between 0 and V_{th} (threshold voltage). It is evident from eq.(2), that subthreshold leakage current depends exponentially on V_{th} . Due to increase in V_{th} , leakage current decreases and vice versa, so to maintain leakage parameter, threshold voltage (V_{th}) has to be reduced.

When the circuit is in operation or active mode, the major concern is run time leakage power (power in active mode). The efficient method to reduce the power in active mode is stacking of transistors. When more than one transistor is in series in CMOS circuits, the leakage current has a strong dependence on a number of turned off transistors. This is known as Stack effect or Self-reverse bias effect [4]. Leakage power depends exponentially on four voltages of transistors (body, gate, source, drain). Fig.3 shows the circuit which consists of a number of transistors.

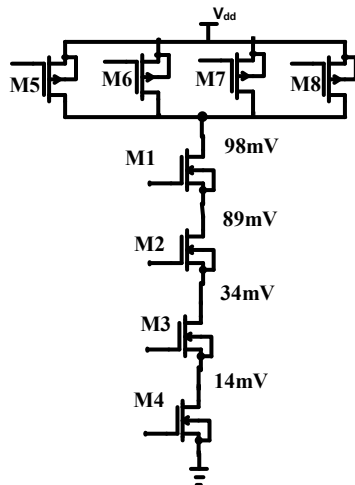


Fig.3. A CMOS Implemented Circuit

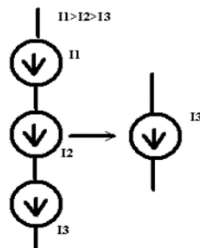


Fig.4. Different current sources in series results in current of smallest magnitude

The circuit shown in Fig.3 has 4 inputs, therefore, it can have 16 different input combinations. Based on the combination there will be a different number of transistors in cascade which are off. Suppose all inputs are logic 0. Since the gate voltage of all nMOS transistors is 0 then they are in the cutoff region. Due to leakage current, there will be a positive voltage at every node in the pull down the network denoted as V_n . There are some important points to understand [4]:

(a). Refer to the M1 transistor from the above diagram, there is a positive voltage at its source and gate is connected to 0 voltage. V_{GS} will be negative and due to the exponential relation between V_{GS} and leakage current, subthreshold leakage current will decrease.

(b). Due to $V_n > 0$, V_{SD} (body to source voltage) will be negative and hence subthreshold leakage current will decrease.

If only one of the transistor in the pull down network is off, then there will be virtually zero voltage at the source of the

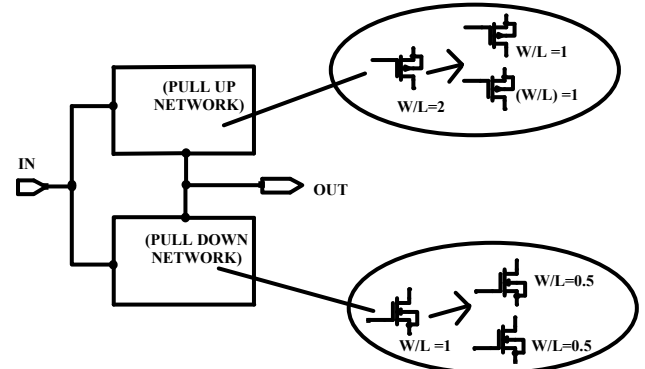


Fig.5. Stacking technique

To reduce the leakage power, replace the transistors in pull-down network and pull up a network with a transistor which has half of the aspect ratio of original transistors [5-8]. While applying the stacking the area of the circuit must not increase i.e. the area occupied by a new group of transistors after stacking must be equal to the area occupied by original transistors before stacking.

B. MTCMOS Technique

Sub-threshold leakage is larger in magnitude than any other leakage at lower technology. Due to modern techniques, VLSI field is going towards deep sub-micron technology where V_{th} is decreasing and delay also decreasing as the delay is directly proportional to the threshold voltage. Since sub-threshold leakage depends exponentially on V_{th} and when V_{th} decreases then there will be a rise in leakage current. Therefore, a balance should be made between high performance and power dissipation because of the following reasons [10]:

(a) Due to low V_{th} , delay will reduce and leakages will increase due to the exponential relation between I_{sub} and V_{th} .

(b) Due to the high value of V_{th} , delay will increase with reduction in leakages.

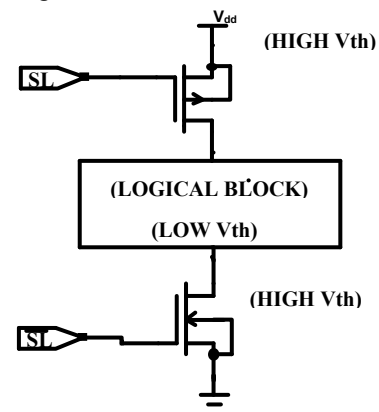
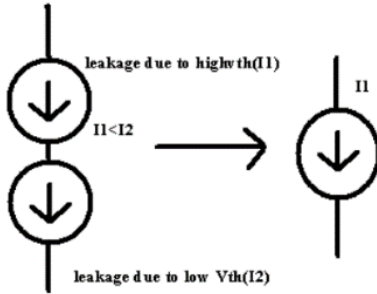


Fig.6. MTCMOS model consisting of low V_{th} and high V_{th} transistors

One of the major challenges is to reduce power dissipation in standby mode and increase performance in active mode. Therefore, to reduce power in active mode well-renowned technology called Multi-threshold CMOS technology (MTCMOS) is used where transistors of different threshold voltages (V_{th}) will be used. Fig.6. shows the MTCMOS technique[5]:

In the MTCMOS model, two sleep transistors (nMOS and pMOS) are used in series with the logical block. Sleep transistors are transistors which are controlled by signals called sleep signals (SL and $\overline{SL}$). Since, subthreshold leakages depend exponentially on V_{th} then based on the status of the sleep transistors circuit will work. In active mode, when both sleep transistors are on then at the top and bottom of the logical block there will be virtual ground and virtual V_{dd} . Then, the logical block will get power supply and based on inputs circuit will give output and leakage current will be small. The circuit will be in standby mode when sleep transistors are in cut-off mode then there will be no virtual power supply for logical block and irrespective of



inputs the leakage current will be less as shown below:

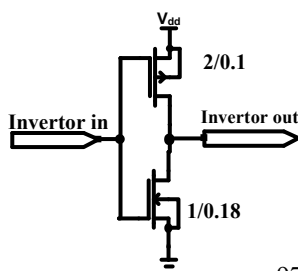
Fig.7. Subthreshold currents in standby mode (I1 due to high V_{th} transistors and I2 due to low V_{th} transistors)

Important thing is to understand that the MTCMOS technique is state destructive technique, i.e., outputs from active mode will not be saved in standby mode.

C. Forced Stack Inverter

The inverter is implemented with forced stacking technique and will be used in the improved circuit [2]. The technology used in simple inverter is 180nm and width of the channel for nMOS is 1μm and for pMOS width of the channel is 2μm.

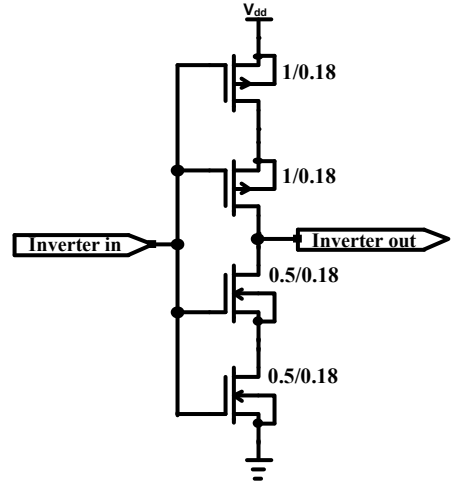
The original inverter circuit consists of two transistors, one nMOS transistor in the pull down the network (PDN) and one pMOS transistor in a pull up the network (PDP). Inverter is implemented with the stacking technique in PDN and PDP where each transistor in the original inverter is replaced by two transistors of half of the aspect ratio of original transistors i.e width of channel for nMOS in the new circuit is 0.5μm and width of channel for pMOS in the new circuit is 1μm and area occupied by the transistor will be the same. This design will help in reducing leakage



current and power dissipation in

Fig.8. Normal Inverter Implementation by CMOS logic

inverter will be low. Fig.9. shows the design of forced stack inverter. Table II shows the comparison of power dissipation for normal inverter and forced stack inverter techniques. Fig.10. shows the graphical comparison of power dissipation



for normal inverter and forced stack inverter.

Fig.9. Forced stack inverter Implementation by CMOS logic

Table II: Comparison of Power Dissipation in Inverter

$V_{dd}(V)$	Inverter(pW)	Forced Stack Inverter(pW)
1	9.968	2.64
1.1	11.781	3
1.2	13.778	3.34
1.3	15.969	3.69
1.4	18.365	4.04
1.5	20.977	4.46
1.6	23.819	4.76
1.7	26.909	5.01
1.8	30.23	5.2
1.9	33.83	5.53

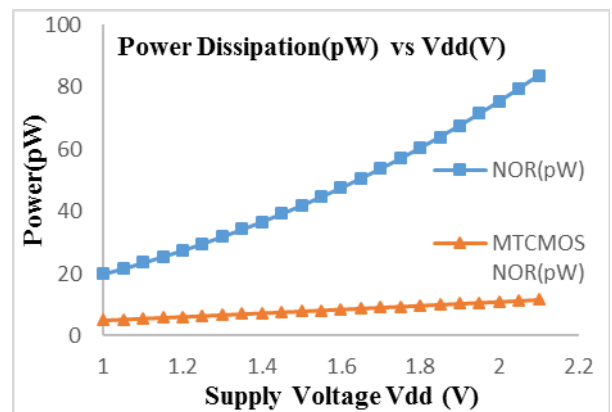


Fig.10. Graphical comparison of power dissipation for inverter and forced stacked inverter

D. MTCMOS NOR

Sleep transistors of high threshold voltage than that of normal transistors are connected in the logic block. One of the sleep transistors is connected between V_{dd} and logic block and another one is connected between the logic block and ground as shown in Fig.11.

In the earlier work in the field of the 1-bit comparator, the NOR gate was implemented by 4 MOSFET (2 nMOS and 2 pMOS). The technology used is 180nm. The NOR model based on MTCMOS technique as described earlier

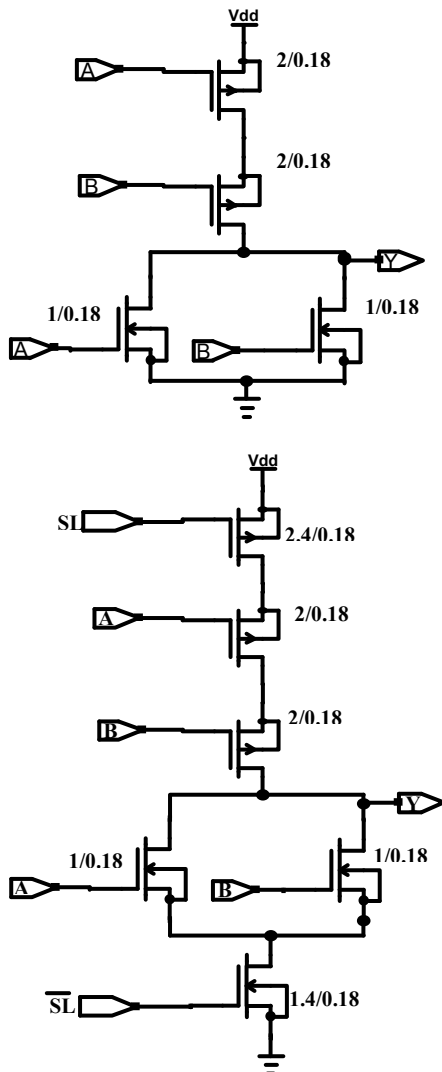


Fig.11. Normal NOR Implementation by CMOS logic

Fig.12. MTCMOS NOR implementation by CMOS logic

Sleep transistors are controlled by sleep signals (SL and $\overline{SL}$). Sleep transistor is switched on when SL becomes 0 and circuit behaves properly but they are in the cut-off region when SL changes to 1 then the circuit goes in standby mode and due to the high threshold voltage, the leakage current will be low and due to that, less power will be dissipated. Table III shows the comparison of power dissipated by NOR and MTCMOS NOR. Fig.13. shows the graphical comparison of power dissipation for NOR and MTCMOS NOR.

Table III: Comparison of Power Dissipation in NOR and MTCMOS NOR

$V_{dd}(V)$	NOR (pW)	NOR Using MTCMOS (pW)
1	19.93	5.01
1.1	23.56	5.59
1.2	27.55	6.16
1.3	31.98	6.75
1.4	36.72	7.32
1.5	41.95	7.92
1.6	47.63	8.52
1.7	53.80	9.13

1.8	60.47	9.74
1.9	67.69	10.36

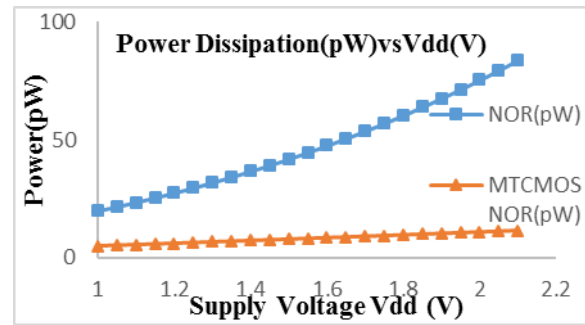


Fig.13. Graphical comparison of power dissipation for normal NOR and MTCMOS NOR

III. IMPROVED CIRCUIT

In this Section, the new comparator is introduced. Fig.14. shows the basic comparator which uses simple logic for comparison. The comparator is improved with the help of stacking technique which results in decrease in power dissipation in the circuit. Fig.15. shows the reported comparator in previous work. The existing comparator has been modified with different techniques and logic. The logic is obtained using Karnaugh maps and truth table of the comparator.

Fig.14. Basic Comparator

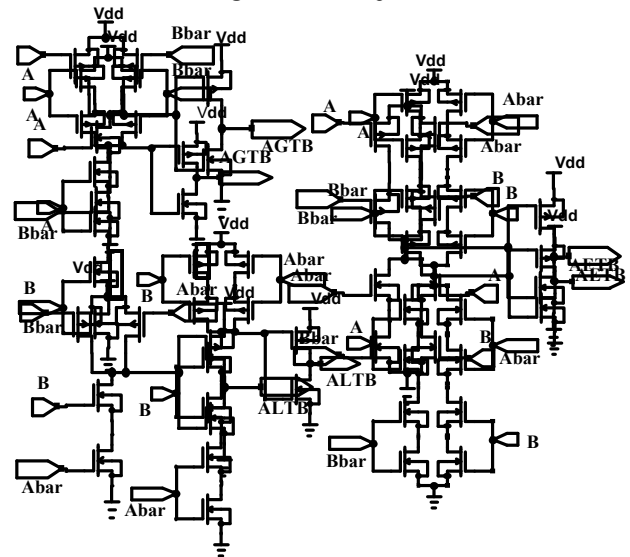


Fig.15. Previous reported comparator [2]

Fig.16. shows the karnaugh maps of the 1-bit comparator from which logical equation of outputs of the comparator is obtained.

	AGTB			ASTB			AETB	
	B'	B		B'	B		B'	B
A'				A'	1		A'	1
A	1			A			A	1

Fig.16. Karnaugh maps for 1-bit Comparator

The techniques which are used in the proposed model is MTCMOS and forced stacking. Forced stacking used in inverter, results in improvement of its performance. Similarly, MTCMOS technique which is implemented in

NOR gate improves the performance of NOR by decreasing power dissipation and delay.

The working of the circuit is the same as a normal circuit which is to compare the two inputs. The new design contains sleep signals which control the sleep transistors. Sleep transistors will be on when the sleep signal turns off i.e. $SL=0$, and they will operate normally and give the result by comparing inputs. Sleep transistors will turn off when sleep signal changes to 1 i.e. $SL=1$ and circuit goes in standby mode, but the state of the circuit will not be retained as MTCMOS is state destructive technique but due to high V_{th} subthreshold leakage will be low. The setup is shown in the Fig.17 where the outputs of AGTB, ASTB are used to produce AETB. It consists of 3 MTCMOS NOR gates, two forced stack inverters to produce the complement of input signals. Fig.17. shows the design of improved comparator. The technology used in the improved circuit is 180 nm. Therefore, the value of length of the channel of all transistors is 0.18 μm . Table IV shows the value of width of the channel of different transistors.

Table IV: Parameters of MOS UNITS

MOS UNITS	Width(μm)
(P1,P4,P7)	2.4
(N1,N4,N7)	1.4
(P2,P3,P5,P6,P8,P9)	2.0
(N2,N3,N5,N6,N8,N9)	1.0
(P10,P11,P12,P13)	1.0
(N10,N11,N12,N13)	0.5

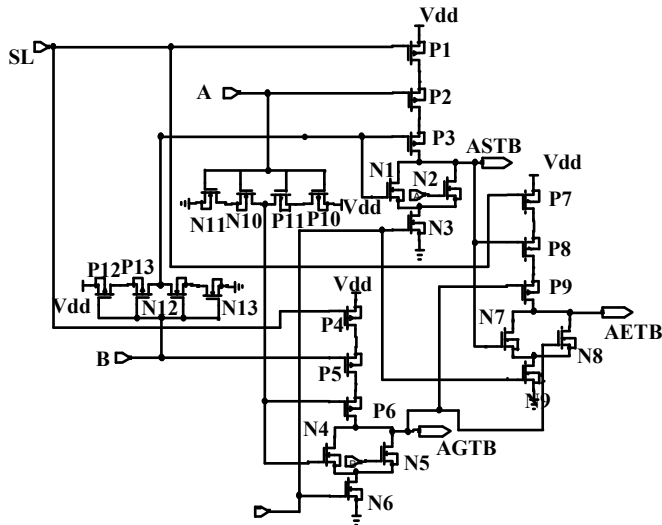
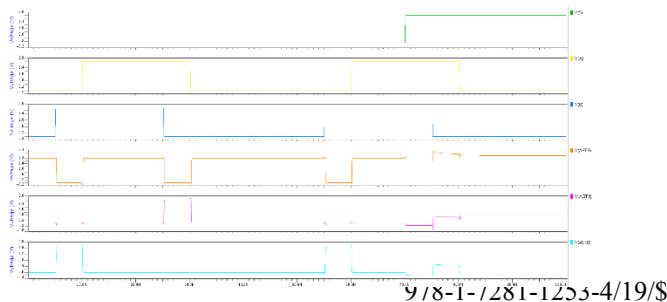


Fig.17. Improved Comparator

IV. SIMULATION AND RESULTS

The simulations of the proposed model shown in Fig.17 are obtained in a 0.18 μm CMOS technology at 300K in SPICE. Supply voltage V_{dd} is varied from 1V to 2.2V. Input signals of frequency 20MHz are used in the improved design. Sleep signals SL and $\overline{SL}$ of frequency 25MHz are



used in the design. The output is shown in Fig 18.

Fig.18. Output of Proposed Comparator

Table V shows the simulation results of power dissipation by the proposed comparator and the delay introduced by the circuit. The power is calculated in SPICE software.

Table V: Power dissipation and Delay in Proposed Model

$V_{dd}(V)$	Power(pW)	Delay(pS)
1	20.20	442.47
1.1	23.15	336.56
1.2	26.25	316.47
1.3	29.51	304.57
1.4	32.09	300.83
1.5	36.66	294.48
1.6	40.17	264.54
1.7	44.06	311.75
1.8	48.12	330.50
1.9	52.37	361.61
2.0	56.8	443.47
2.1	61.42	593.94

For comparison, results were compared with earlier models which are shown in Table VI and Table VII.

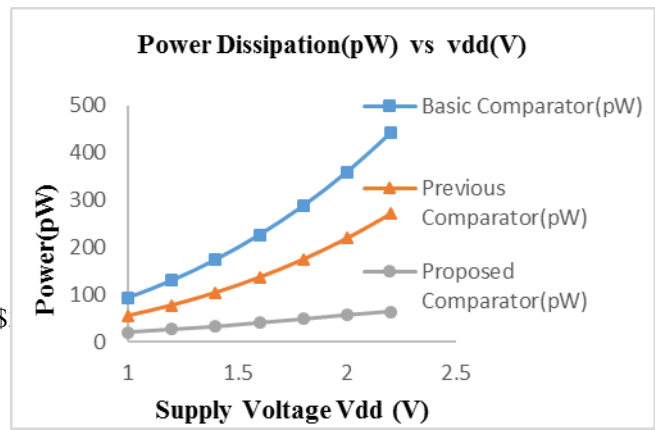
Table VI: Comparison of power dissipation for comparators

$V_{dd}(V)$	Basic Comparator (pW)	Previous Comparator (pW)	Proposed Comparator (pW)
1	93.06	54.58	20.20
1.2	129.84	76.9	26.25
1.4	174.05	103.92	32.09
1.6	226.45	136.21	40.17
1.8	287.93	174.35	48.12
2.0	359.46	219.00	56.80

Table VII: Comparison of Delay for Comparators

$V_{dd}(V)$	Basic Comparator (nS)	Previous Comparator (nS)	Proposed Comparator (nS)
1	49.21	48.54	0.442
1.2	49.45	48.98	0.316
1.4	49.56	49.19	0.300
1.6	49.63	49.34	0.264
1.8	49.67	49.38	0.330
2.0	49.70	49.47	0.443

Analysis of Table VI proves a reduction in power dissipation is achieved by the proposed model. The power dissipated by the previous model at 1.8V is 174.35pW, at same supply voltage power dissipated by the proposed model is 48.12pS which is approximately 27.6% of the previous one i.e. 72.4% of power dissipated is reduced as compared to the previous comparator. Fig.19. shows the graph which compares the power dissipation for all the comparators. The delay obtained in the proposed design is very less as



compared to previous models which shows the significant improvement of the performance of the circuit as it becomes fast. Fig.20. shows the graph which represents the delay for improved comparator.

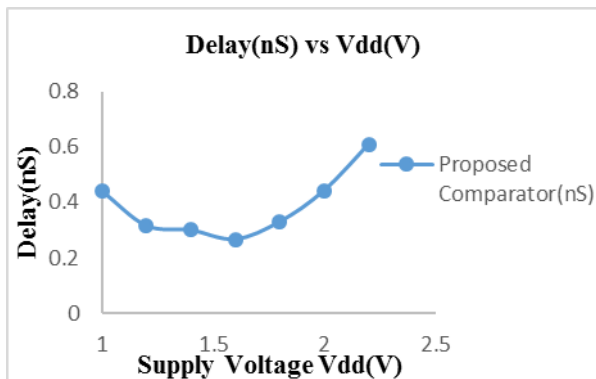


Fig.19. Graphical comparison of power dissipation for Comparators

Fig.20. Graphical representation of delay for Proposed model

The power delay product(PDP) is calculated for V_{dd} from 1V to 2V as shown in TableVIII.

Table VIII: Power delay product for Proposed Model

$V_{dd}(V)$	PDP ($10^{-12}J$)
1.00	8.9
1.10	7.7
1.20	8.3
1.30	8.9
1.40	9.6
1.50	9.6
1.60	10.7
1.70	13.7
1.80	15.9
1.90	18.9
2.00	25.1

Table IX: Comparison of power and delay for all models

Approach Type	Power	Delay
Basic Comparator	287.93(pW)	49.67(ns)
Previous Comparator	174.35(pW)	49.38 (ns)
Proposed Comparator	48.12(pW)	330.61(ps)

V. CONCLUSION

In the proposed work, a new CMOS nm technology-based digital circuit of one-bit comparator has been presented. The proposed circuit has a very low power loss or dissipation (static and dynamic both) and propagation delay as well. In this work, forced stack technique is used in the Inverter circuit and MTCMOS technique in NOR circuit and using k maps and truth tables to improve the design. MOS units are reduced by a large number and thus the power and delay have reduced to a great extent. The proposed circuit

consumes 48.12(pW) of power. The delay in proposed circuit is 330.12 (pS). Using logic and K-maps equation the number of MOS units are reduced by 14 in number and resulting in improvement in power and delay. From the above results, it is clearly observed that the power and propagation delay has reduced considerably.

VI. REFERENCES

- [1] E. R. Thuraka, "High Performance Arithmetic and Logic Unit with Enhanced MTCMOS and Transistor Stacking Techniques", *ajct*, vol. 4, no. 1, Apr. 2018
- [2] A. Kumar and M. Kumar, "Improved design of CMOS 1-bit comparator with stacking technique," 2017 2nd International Conference on Telecommunication and Networks (TEL-NET), Noida, 2017, pp. 1-5. doi: 10.1109/TEL-NET.2017.8343537
- [3] Pushpa Saini, Rajesh Mehra, "Leakage Power Reduction in CMOS VLSI Circuits", *International Journal of Computer Applications*, vol. 55, issue 8, pp. 42-48 [3] Ankita Nagar, Vidhu Parmar, "Implementation of Transistor Stacking Technique in Combinational Circuits", *IOSR Journal of VLSI and Signal Processing (IOSR-JVSP)* vol. 4, Issue 5, Ver. I (Sep-Oct 2014), PP 01-05 e-ISSN: 2319 - 4200, p-ISSN no. 319 - 4197 www.iosrjournals.org
- [4] Neha Goyal, Renu Singla, Puneet Goyal, "Study and Analysis of Universal Gates Using Stacking Low Power Technique", Neha Goyal et al, / (IJCST) *International Journal of Computer Science and Information Technologies*, vol. 5 (3), 2014, 4200-4204
- [5] Gholamreza Karimi, Adel Almoradi, "Multi-Purpose Technique to Decrease Leakage Power in VLSI Circuits," *Canadian Journal on Electrical and Electronics Engineering* vol. 2, no. 3, Mar. 2011
- [6] Shaik Nazia Tarannum, M. Raju, B. Ram Prasad, "Design of High Speed and Power Efficient Full Subtractor Using MTCMOS," *International Journal of Research in Engineering, Science and Management* ISSN: 2581-5792, vol. 1, Issue-11, Nov. 2018.
- [7] B. Murmann and B. E. Boser, "A 12-bit 75-MS/s pipelined ADC using open-loop residue amplification," *IEEE J. Solid-State Circuits*, vol. 38, no. 12, pp. 2040-2050, Dec. 2003.
- [8] B. M. Dharmendra, B. Arun, B. Nandurbarkar, "Study and Implementation of Comparator in CMOS 50 NM Technology" *International Journal of Research in Engineering and Technology (IJRET)*, ISSN: 2321-7700, Issue-2, pp. 252-255, Feb. 2014.
- [9] M.E. Johnson, D. Somasekhar, and K. Roy, "Models and Algorithms for Bounds on Leakage in CMOS Circuits," *IEEE Transactions on Computer-Aided Design on Integrated Circuits and Systems*, vol. 18, no. 6, pp. 714-725, Jun. 1999.
- [10] H. E. Neil, Weste, D. Harris and A. Banerjee, "CMOS VLSI Design," Pearson Education, Third Edition, pp. 9-10, 2008.
- [11] M. Hassan, R. Mehra, "Design Analysis of 1-bit CMOS comparator," *International Journal of Scientific Research Engineering & Technology (IJSRET)* ISSN: 2278-0882, pp. 68-72, Mar. 2015
- [12] J.C. Park, V. J. Mooney III and P. Pfeifferberger, "Sleepy Stack Reduction of Leakage Power," *Proceeding of the International Workshop on Power and Timing Modeling, Optimization and Simulation*, pp. 148-158, Sept. 2004.

- [13] S. Kang and Y. Leblebici "CMOS Digital Integrated Circuit, Analysis and Design" (Tata McGraw-Hill, 3rd Ed, 2003).
- [14] I.C. Park, V. I. Mooney III and P. Pfeifferberger, "Sleepy Stack Reduction of Leakage Power," in Proc. 2004 International Workshop on Power and Timing Modeling, Optimization and Simulation, pp. 148-158, September 2004.
- [15] M.E. Johnson, D. Somasekhar, and K. Roy, "Models and Algorithms for Bounds on Leakage in CMOS Circuits," IEEE Transactions on Computer Aided Design on Integrated Circuits and Systems, vol. 18, no. 6, pp. 714-725, Jun. 1999.
- [16] Adel S. Sedra, Kenneth C. Smith: Book Microelectronic Circuits: Theory and Applications Seventh Edition.